

TPS54202H (main +14V)

<https://www.ti.com/lit/ds/symlink/tps54202h.pdf?ts=1771882131912>

Setting the output voltage (E96):

$V_{OUT} = V_{ref} \times \left[\frac{R2}{R3} + 1 \right]$	R3 11 kR	R2 249 kR	V_{ref} 0,596 V	V_{OUT} 14,1 V
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Setting under voltage-lockout:

$R4 = \left(\frac{V_{OUT(START)} - V_{START} - V_{STOP}}{V_{ENRISING}} \right) / I_h$	V_{ENfalling} 1,25 V	V_{ENrising} 1,28 V	V_{START} 15 V	V_{STOP} 14 V	I_h 1 μA	R4 648,4 kR	649kR E96
$R5 = \frac{R4 \times Rpd}{\left(\frac{V_{START}}{V_{ENRISING}} - 1 \right) \times Rpd - R4}$	R4 648,4 kR	V_{ENrising} 1,28 V	V_{START} 15 V	Rpd 1 MR		R5 64,4 kR	64,9kR E96

Setting the minimum value of the output inductor

$L_{MIN} = \frac{V_{OUT} \times (V_{IN(MAX)} - V_{OUT})}{V_{IN(MAX)} \times K_{IND} \times I_{OUT} \times f_{sw}}$	V_{OUT} 14,1 V	V_{IN(MAX)} 24 V	K_{IND} 0,3	I_{OUT} 0,85 A	f_{sw} 500 kHz	L_{MIN} 45,6 μH	47 μH E12
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Setting the RMS inductor current

$I_{L(RMS)} = \sqrt{\left(I_{OUT(MAX)}^2 \times \frac{1}{12} \right) + \left(\frac{V_{OUT} \times (V_{IN(MAX)} - V_{OUT})}{V_{IN(MAX)} \times L_O \times f_{sw} \times 0,8} \right)^2}$	V_{OUT} 14,1 V	V_{IN(MAX)} 24 V	L_O 47 μH	I_{OUT(MAX)} 1,2 A	f_{sw} 500 kHz	I_{L(MAX)} 1,20 A
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Setting the peak inductor current

$I_{L(PK)} = I_{OUT(MAX)} + \frac{V_{OUT} \times (V_{IN(MAX)} - V_{OUT})}{1,8 \times V_{IN(MAX)} \times L_O \times f_{sw}}$	V_{OUT} 14,1 V	V_{IN(MAX)} 24 V	L_O 47 μH	I_{OUT(MAX)} 1,2 A	f_{sw} 500 kHz	I_{L(PK)} 1,35 A
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TPS54202H (+7V)

<https://www.ti.com/lit/ds/symlink/tps54202h.pdf?ts=1771882131912>

Setting the output voltage (E96):

$V_{OUT} = V_{ref} \times \left[\frac{R2}{R3} + 1 \right]$	R3 15 kR	R2 165 kR	V_{ref} 0,596 V	V_{OUT} 6,98 V
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Setting under voltage-lockout:

$R4 = \left(\frac{V_{OUT(START)} - V_{START} - V_{STOP}}{V_{ENRISING}} \right) / I_h$	V_{ENfalling} 1,25 V	V_{ENrising} 1,28 V	V_{START} 14 V	V_{STOP} 13 V	I_h 1 μA	R4 671,9 kR	681kR E96
$R5 = \frac{R4 \times Rpd}{\left(\frac{V_{START}}{V_{ENRISING}} - 1 \right) \times Rpd - R4}$	R4 681 kR	V_{ENrising} 1,28 V	V_{START} 14 V	Rpd 1 MR		R5 73,6 kR	73,2kR E96

Setting the minimum value of the output inductor

$L_{MIN} = \frac{V_{OUT} \times (V_{IN(MAX)} - V_{OUT})}{V_{IN(MAX)} \times K_{IND} \times I_{OUT} \times f_{sw}}$	V_{OUT} 6,98 V	V_{IN(MAX)} 24 V	K_{IND} 0,3	I_{OUT} 0,15 A	f_{sw} 500 kHz	L_{MIN} 220 μH	exact E12
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Setting the RMS inductor current

$I_{L(RMS)} = \sqrt{\left(I_{OUT(MAX)}^2 \times \frac{1}{12} \right) + \left(\frac{V_{OUT} \times (V_{IN(MAX)} - V_{OUT})}{V_{IN(MAX)} \times L_O \times f_{sw} \times 0,8} \right)^2}$	V_{OUT} 6,98 V	V_{IN(MAX)} 24 V	L_O 220 μH	I_{OUT(MAX)} 0,3 A	f_{sw} 500 kHz	I_{L(MAX)} 0,30 A
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Setting the peak inductor current

$I_{L(PK)} = I_{OUT(MAX)} + \frac{V_{OUT} \times (V_{IN(MAX)} - V_{OUT})}{1,8 \times V_{IN(MAX)} \times L_O \times f_{sw}}$	V_{OUT} 6,98 V	V_{IN(MAX)} 24 V	L_O 220 μH	I_{OUT(MAX)} 0,3 A	f_{sw} 500 kHz	I_{L(PK)} 0,33 A
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LT3471 (±18V)

<https://www.mouser.pl/datasheet/3/1014/1/3471fb.pdf>

Setting the output voltage (E96):

$V_{OUT} = V_{FBP} \left(1 + \frac{R1}{R2} \right)$	R1 255 kR	R2 15 kR	V_{FBP} 1 V	V_{OUT} 18 V
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Setting the output voltage (E96):

$V_{OUT} = V_{REF} \left(\frac{R1}{R2} \right)$	R1 274 kR	R2 15 kR	V_{REF} 1 V	V_{OUT} 18,3 V	/ negative
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